

Strategic Sourcing & Drilling Bit Performance — Executive Summary

1. Overall Performance

The analysis shows significant variation in drilling-bit performance across suppliers.

The highest observed **Average Bit Life / Average Footage per Bit** was:

Supplier	Average Bit Life (m/bit)	Bits
DORADO	7,867.73	3
CAT	4,285.42	5
DRILLCO	3,794.01	60
AP DRILLING	3,690.57	333
EPIROC	3,095.15	59
TERELION	1,025.30	19
MINCON	634.29	9

DORADO shows the highest average bit life, while MINCON shows the lowest.

However, DORADO and CAT have very small sample sizes, so their results should be interpreted cautiously.

2. Price vs. Performance

The analysis does **not show a clear relationship between bit price and average footage per bit**.

Examples from the data:

- A price of 2,775 *corresponds to approximately 4,285m/bit*. — A price of 3,690 corresponds to approximately 3,095 m/bit.
- A price of 3,990 *corresponds to 7,868m/bit*, but this is based on only 3 bits. — 5,760 corresponds to only 1,451 m/bit.
- A price of \$3,750 corresponds to only 634 m/bit.

Therefore, the data does **not support the hypothesis that a higher-priced bit automatically provides better performance**.

Key sourcing insight: Price should not be used as a proxy for product performance. Supplier evaluation should instead consider PDC, bit life, failure rate and, when available, TDC/ROP.

3. Premature Failure Analysis

The overall premature failure rate is:

$$152 / 488 = 31.15\%$$

This means approximately **1 out of every 3 recorded bits** was classified as a premature failure under the current failure-category mapping.

Main Failure Modes

Failure Category	Bits	Share
Normal Wear / Other	336	68.85%
Stuck Bits	110	22.54%
Broken Inserts	39	7.99%

The two identified premature-failure categories account for approximately **30.53%** of the observations.

The high incidence of **Stuck Bits** requires further investigation. However, it should not automatically be interpreted as a manufacturing defect. Potential causes may include:

- Drilling parameters
- Rock formation
- Equipment conditions
- Operator practices
- Bit selection
- Operating environment

Therefore, supplier attribution should be validated using additional operational data.

4. Suppliers with Higher Premature Failure Rates

The suppliers with the highest observed premature-failure rates are:

Supplier	Bits Consumed	Premature Failures	Failure Rate
TERELION	19	19	100.00%
MINCON	9	9	100.00%
DRILLCO	60	27	45.00%

These results require further investigation, particularly because sample sizes differ significantly between suppliers.

A 100% failure rate based on 9 bits should not be interpreted in the same way as a 45% failure rate based on 60 bits without considering statistical uncertainty and operating conditions.

5. Mine-Level Findings

The analysis identifies **BVC** as having the highest observed premature-failure incidence.

Mine	Bits Consumed	Premature Failures	Failure Rate
BVC	311	151	48.55%

This suggests that **mine-specific operating conditions may be an important factor** in drilling-bit performance.

The next analysis should therefore compare suppliers under equivalent conditions:

Mine → Equipment → Bit Size → Supplier

This will help distinguish supplier/product effects from operational effects.

6. Bit Size Analysis

The **10.625-inch bits** show particularly high failure rates in several supplier/mine combinations.

Supplier	Mine	Bit Size	Bits	Premature Failures	Failure Rate
AP DRILLING	BVC	10.625"	2	2	100.00%
MINCON	BVC	10.625"	9	9	100.00%
TERELION	BVC	10.625"	9	9	100.00%

For 12.25-inch bits:

Supplier	Mine	Bits	Premature Failures	Failure Rate
AP DRILLING	BVC	213	92	43.19%
DRILLCO	BVC	60	27	45.00%
DORADO	BVC	3	1	33.33%
CAT	BVC	5	1	20.00%
EPIROC	MDC	59	1	1.69%
AP DRILLING	MDC	118	0	0.00%

A notable observation is the difference between **BVC and MDC for the same supplier**. For example, AP DRILLING has a 43.19% premature-failure rate for 12.25-inch bits in BVC compared with 0% in MDC.

This reinforces the need to investigate **mine-specific operating conditions before attributing failures exclusively to suppliers**.

7. Supplier Portfolio Exposure

AP DRILLING dominates the current portfolio:

- **70.6% of total spend**
- **71.9% of total footage**
- **68.2% of total bit consumption**

This makes AP DRILLING the supplier with the largest financial and operational exposure in the current dataset.

A sourcing strategy should therefore evaluate both **supplier performance** and **portfolio exposure**.

A relatively small improvement in PDC, bit life or failure rate for a high-volume supplier could have a significant impact on total drilling economics.

8. Supplier Performance Overview

The strongest separation between suppliers is observed in **Average Bit Life and PDC**.

Supplier	Avg. Bit Life (m/bit)	PDC
DORADO	7,867.73	0.507
CAT	4,285.42	0.648
DRILLCO	3,794.01	1.048
AP DRILLING	3,690.57	1.189
EPIROC	3,095.15	1.192
TERELION	1,025.30	4.712
MINCON	634.29	5.912

The results show substantial differences in both cost and bit life.

However, supplier comparisons should be interpreted together with:

- Sample size
- Mine
- Equipment
- Bit size
- Failure mode
- Operating conditions

Answers to the Key Questions

1. Which supplier has the lowest cost per meter?

Using:

PDC = Total Drill Bit Expenditure / Total Footage

the supplier PDC results are:

Supplier	PDC
DORADO	0.507
CAT	0.648

Supplier	PDC
DRILLCO	1.048
AP DRILLING	1.189
EPIROC	1.192
TERELION	4.712
MINCON	5.912

Answer

DORADO has the lowest PDC at 0.507 per meter.

However, this result is based on only **3 bits**, so it should be considered an initial observation rather than a definitive sourcing conclusion.

2. Which supplier has the lowest TDC?

TDC cannot currently be determined from the available dataset.

The required formula is:

$$\text{TDC} = \text{PDC} + (\text{Rig Rate} / \text{ROP})$$

The current dataset does not contain:

- **ROP (Rate of Penetration)**
- **Rig Rate**

Therefore, calculating or ranking TDC at this stage would require assumptions that are not supported by the source data.

Answer

TDC cannot be reliably calculated with the current dataset.

The next data integration stage should incorporate **ROP and Rig Rate** to enable a complete TDC analysis.

3. Is there a relationship between price and performance?

Answer

No clear positive relationship was identified between bit price and performance.

The data contains examples where:

- Lower-priced bits achieved relatively high footage.
- Higher-priced bits achieved relatively low footage.

- Some high-priced observations achieved high footage but were based on very small sample sizes.

Therefore:

The current data does not support the hypothesis that higher bit price necessarily results in better drilling performance.

A more rigorous analysis should control for:

Supplier + Mine + Equipment + Bit Size + Operating Conditions

before making a final conclusion about the relationship between price and performance.

Final Executive Conclusion

The analysis identifies **significant variation in drilling-bit economics and reliability across suppliers**. DORADO currently presents the lowest PDC and highest average bit life, while MINCON and TERELION show substantially higher PDC and lower bit life. However, the small sample sizes of some suppliers limit the reliability of direct comparisons.

AP DRILLING represents the largest portfolio exposure, accounting for approximately 70% of spend and 72% of total footage. Consequently, its performance represents an important opportunity for cost and operational optimization.

The overall premature-failure rate of **31.15%** is driven primarily by **Stuck Bits (22.54%)** and **Broken Inserts (7.99%)**. BVC shows a particularly high observed premature-failure rate of **48.55%**, suggesting that mine-specific operating conditions may play an important role.

Finally, the analysis **does not identify a clear relationship between price and performance**, indicating that supplier selection should not be based on unit price alone.

The recommended next step is a controlled comparison of:

Supplier × Mine × Equipment × Bit Size × Failure Mode × Performance

combined with **ROP and Rig Rate** to calculate TDC and establish a more complete total-cost-of-ownership view.

Comienza a programar o [generar](#) con IA.

